

Bounded-Authority Verification for Safety-Critical Pesticide Advisory in Low-Resource Bengali Agriculture

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ARTICLE INFO

Keywords:

Agricultural decision support
Large language models
Retrieval-augmented generation
Safety-critical verification
Bounded factual authority
Selective resolution
Low-resource NLP
Edge computing

ABSTRACT

Large Language Models and standard Retrieval-Augmented Generation (RAG) pipelines are fluent in agricultural extension but lack factual grounding guarantees. In crop protection, a fluent sentence can silently misbind a pesticide to the wrong crop, inflate a dosage, or drop a mandatory Pre-Harvest Interval (PHI), producing failures that remain confident-sounding and dangerous. We present the **Bounded-Authority Advisory Architecture (BAA)**, which separates factual authority from linguistic realization: the generative model may phrase and explain, but it cannot invent or mutate safety-critical agronomic values. BAA enforces an 11-slot, single-record, fail-closed certification contract and selectively routes queries among deterministic resolution, grounded generation, clarification, abstention, and escalation.

On 10,000 adversarial relational-misbinding attacks, BAA rejects 100.0% of corrupted chemical claims (95% Wilson CI [99.96, 100.0]), against 80.0% dangerous acceptance for a lexical baseline and 38.65% for an LLM-as-a-judge verifier. Calibrated abstention sustains 84.56% advisory coverage at a 1.26% selective-risk level across 20,112 held-out cases. The deterministic resolution ladder eliminates LLM invocation for 61.5% of queries, cutting weighted latency 2.28× and per-query cost by 89.7%. The results demonstrate that subordinating generative language to bounded factual authority yields a dependable, auditable, and affordable foundation for safety-critical agricultural decision support.

1. Introduction

1.1. Agricultural Advisory as a Safety-Critical Domain

Agricultural crop protection is an inherently safety-critical domain. A pesticide recommendation is not a passage of prose but a tightly coupled tuple of decision variables: the host crop, the target pathogen, the registered active ingredient, its formulation code, a bounded dosage interval with explicit units, the solvent volume, the spray interval, and the mandatory pre-harvest interval (PHI). Consider late blight management in potato: the correct advisory specifies mancozeb 80 WP at 2.0–2.5 g per litre of water, applied at seven-day intervals with a 14-day PHI. Altering any single field—substituting the crop, inflating the dose, or omitting the PHI—produces a superficially valid recommendation that can cause phytotoxicity, dangerous chemical residues on harvested produce, environmental contamination, or acute poisoning of the applicator. The consequence of an error is not a degraded user experience; it is a measurable harm to a farmer, a field, or a food supply chain.

Large Language Models (LLMs) generate fluent, plausible-sounding text across all of these fields, and Retrieval-Augmented Generation (RAG) pipelines can anchor that fluency to retrieved documents [10, 12]. Yet neither fluency nor the presence of relevant text in a retrieval window determines the correctness of a safety-critical agronomic value. In unconstrained generative systems and in standard

RAG deployments, factual authority rests implicitly with the model’s parametric memory and with whatever passages happen to be retrieved. When retrieved documents individually contain valid facts drawn from different records, the generator assembles a semantically coherent composite—the correct crop paired with a dosage taken from an unrelated pathogen’s entry, or the correct active ingredient coupled with a PHI lifted from a different formulation.

1.2. Retrieval Failure Modes in Agricultural RAG

Retrieval quality in agricultural domains varies sharply with query type and language condition. Text-first retrieval Hit@1 on standard formal Bengali reaches approximately 72–73%, but falls to 44.7–45.7% on regional dialects and 41.5–41.6% on romanized Banglish input (Layer E17, threshold sweep at $\theta=0.7$), leaving the generator to operate on degraded or absent evidence. These failures are the dominant failure mode, not rare edge cases. In a 100-case root-cause audit conducted as part of this work (Layer E10), retrieval evidence omission accounted for 28% of advisory failures and generative relational misbinding for 22%; both failure modes produced outputs that remained fluent and confident, which is precisely what makes them dangerous.

1.3. The Missing Authority Boundary

Existing agricultural advisory systems can be categorized into two paradigms. Generative conversational systems such as Farmer.Chat [12] and KrishokBondhu [1] treat the LLM as the end-to-end reasoning engine, prioritising multilingual fluency and conversational usability for smallholder farmers. Structured knowledge systems—decision tables, knowledge graphs, and crop simulation

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wrappers [6, 7]—hold curated facts but cannot converse with farmers in natural language. A growing body of work addresses individual reliability properties: temporal validity [9], multimodal disease diagnosis [4], post-generation faithfulness assessment [5], and adversarial robustness of retrieval pipelines [8]. What remains absent is an evaluated architecture that makes the authority relationship between deterministic agricultural facts and generative language explicit, and that quantifies the safety–coverage–latency frontier of that relationship under adversarial, linguistic, and operational stress.

1.4. Proposed Solution: Bounded-Authority Advisory Architecture

We propose the **Bounded-Authority Advisory Architecture (BAA)**, a decision-support framework for safety-critical crop protection in which factual authority is structurally separated from linguistic realization. The generative model retains its role as a phrasing and explanation engine, but it cannot invent, mutate, or recombine safety-critical agronomic values. Every advisory is certified against an 11-slot, single-record, fail-closed contract: all contract fields must be jointly satisfied by one authoritative evidence record, and any failure triggers deterministic routing to clarification, safe abstention, or escalation to the national Krishi Call Center (dial 16123).

Design Rule. *In safety-critical agricultural advisory, the component that generates fluent language must not be the component that holds factual authority.*

This rule has a concrete operational consequence: a failed safety-critical evidence condition is never repaired by generating additional text. If the 11-slot contract cannot be jointly satisfied from a single accredited record, the system does not attempt a best-effort completion; it abstains or escalates. Throughout this paper, **BAA** refers to the architecture; **KrishokChat** refers to the deployed system artifact.

1.5. Research Questions and Contributions

Table 1 summarizes the five governing research questions, primary metrics, and corresponding result sections.

The contributions of this work are threefold:

1. **Bounded-authority advisory architecture.** A decision-support architecture that separates factual authority from generative language through an 11-slot, fail-closed, single-record certification contract (Sections 3 and 4).
2. **Risk- and evidence-aware selective resolution policy.** A five-tier resolution ladder that routes each query among deterministic resolution, grounded generation, clarification, abstention, and escalation, with a frozen threshold validated on held-out data (Sections 3 and 9).
3. **Multi-axis empirical evaluation.** An extensive evaluation covering adversarial safety, empirical advisory

Table 1

Research questions, key metrics, and result sections.

RQ	Question	Key Metric	Section
RQ1	Does BAA certify correct, safe advice on naturalistic and adversarial queries?	CAC, CUAR	\$7
RQ2	Does single-record joint binding protect against misbinding, counterfactual corruption, and evidence conflict?	CUAR, Rejection Rate	\$8
RQ3	Can calibrated abstention trade coverage for a user-specified risk bound?	AURC, Coverage@Risk	\$9
RQ4	Does the architecture preserve safety under linguistic register, temporal, and multimodal uncertainty?	Dialect Coverage, Gazette Adherence	\$9
RQ5	What are the latency, cost, connectivity, and channel-fidelity properties of the constrained pathways?	Latency, Zero-LLM%, Cost/1k	\$10

correctness, risk–coverage calibration, linguistic and temporal robustness, prompt injection security, and constrained-deployment economics, spanning 27 experimental layers and over 60,000 evaluation cases (Sections 6–10).

The remainder of this paper is organized as follows. Section 2 positions the work against the literature. Section 3 formalizes the problem and the certification contract. Section 4 describes the architecture, and Section 5 details knowledge governance. Section 6 presents the methodology. Sections 7–10 report results organized by research question. Sections 11–15 present the discussion, limitations, reproducibility, and conclusion.

2. Related Work

Table 2 provides a systematic positioning of the present work against representative systems across agricultural conversational AI, structured knowledge systems, and selective prediction. The comparison reveals that while individual properties—multilingual fluency, structured facts, temporal awareness, multimodal conditioning, offline operation, and fail-closed safety—have each been demonstrated in isolation, no evaluated system binds a safety-critical advisory contract to a single authoritative record and selectively resolves across deterministic, generative, and abstention paths.

2.1. Agricultural Conversational AI and Structured Knowledge Systems

The deployment of LLM-based agricultural extension has accelerated in recent years. Farmer.Chat [12] demonstrated scalable multilingual conversational advisory serving smallholder farmers across four countries, establishing that LLM-driven interfaces are usable in low-resource agricultural settings. KrishokBondhu [1] introduced a Bengali voice-enabled retrieval pipeline for crop advisory. Both systems treat the LLM as the primary reasoning engine: the model retrieves, synthesises, and delivers the advisory in a

Table 2

Systematic positioning of KrishokChat alongside contemporary agricultural AI, retrieval-augmented decision support, and low-resource advisory systems.

System / Framework	Language	Domain	RAG	Structured Facts	Temporal	Multimodal	Fail-Closed	Edge/Offline
Farmer.Chat [12]	Multilingual	General Agri	✓	Partial	×	×	×	×
KrishokBondhu [1]	Bengali	Crop Advisory	✓	×	×	Voice	×	×
Domain-First RAG [6]	German/English	Livestock	✓	✓	×	×	×	×
TARAG [9]	Chinese/English	Crop Diseases	✓	×	✓	×	×	×
SMART [4]	English/Hindi	Plant Pathology	✓	✓	×	✓	HITL	×
AgroTutor [11]	Spanish	Crop Planning	×	✓	×	×	×	✓
KrishokChat (This Work)	Bengali	Crop Advisory	✓	✓	✓	✓	✓	✓

single generative pass. Under dialectal variation and multi-document retrieval, this design leaves the system exposed to the silent factual corruption described in Section 1.

A complementary line of work embeds structured agricultural knowledge directly into decision support. Han et al. [6] textualised decision trees and tables for a goat-farming knowledge assistant, improving answer consistency over heterogeneous livestock knowledge. TARAG [9] contributed a time-aware knowledge base with time-sensitive re-ranking so that crop–pest recommendations respect phenology and pest life cycles. SMART [4] integrated structured multimodal embeddings with human-in-the-loop verification for plant disease diagnosis. DSSAT-LM [7] coupled LLMs with mechanistic crop simulation by parsing free-text queries into structured simulation parameters. AgroTutor [11] provided offline mobile decision support for data-scarce environments. These systems demonstrate that structured representations, temporal awareness, and domain-specific constraints improve advisory quality. BAA incorporates the same mechanisms—typed facts, temporal validity tags, and multimodal conditioning—but enforces them as binding constraints of a certification contract rather than as retrieval features or post-hoc filters. The distinction is architectural: in BAA, a retrieved fact that fails the single-record joint entailment predicate cannot be certified regardless of its individual plausibility.

2.2. RAG Faithfulness, Verification, and Adversarial Robustness

Faithfulness of generated answers to retrieved evidence has become a first-class evaluation target in the RAG literature. RAGChecker [10] argued that retrieval and generation must be diagnosed separately, introducing fine-grained metrics for each component. Generate-but-Verify [5] formalised post-generation faithfulness prediction as an explicit task with precision and recall variants, showing that better faithfulness assessment improves downstream RAG performance. FaithfulRAG [14] modelled fact-level conflict between retrieved knowledge and parametric memory, and claim-level provenance tracing was demonstrated in scientific verification by SciFact [13].

On the security side, SafeRAG [8] demonstrated that manipulated or poisoned retrieved knowledge can defeat otherwise strong RAG pipelines, highlighting the vulnerability

of retrieval-grounded systems to adversarial evidence injection. These contributions collectively establish that post-hoc textual verification and retrieval-side filtering are necessary but insufficient for safety-critical fields, because they verify the surface form of language rather than the relational authority underlying an advisory. A sentence can be faithful to its retrieved passages in a textual sense while assembling fields from different records into a recommendation that no single authoritative source supports. BAA addresses this gap by certifying an 11-slot tuple against a single evidence record hash, so a faithful-sounding sentence whose fields originate from different records fails certification by construction.

2.3. Selective Prediction, Abstention, and Risk-Aware Resolution

Selective prediction and calibrated abstention are well-established tools in classification, where risk–coverage curves allow a system to trade coverage for a bounded error rate. In conversational and generative settings, abstention has increasingly been treated as a first-class output rather than a failure mode [5]. Within agricultural advisory specifically, the consequences of an incorrect answer are asymmetric: a wrong pesticide recommendation can cause direct harm, whereas a well-formed abstention with a referral to a human expert carries a bounded and manageable cost. BAA applies calibrated abstention to agricultural advisory: the system abstains when the 11-slot contract cannot be jointly satisfied, and the abstention threshold is frozen on a development split and validated on a held-out test split (Section 9). The selective resolution ladder extends this principle beyond binary answer-or-abstain decisions by introducing intermediate actions—clarification, deterministic resolution, and escalation—each with its own evidence and risk conditions.

2.4. Positioning and Differentiation

The work most closely related to BAA in spirit is the combination of structured knowledge enforcement [6, 7] with selective answering and risk calibration. However, existing systems either enforce structure without conversational capability or provide conversational capability without structural enforcement of factual authority. BAA occupies the intersection: a reliability-oriented agricultural computing architecture in which factual authority is explicitly bounded, uncertainty terminates safely through calibrated abstention or escalation, and generative language is

subordinate to validated evidence. The novelty resides not in any individual component—structured knowledge bases, retrieval pipelines, LLM generation, and abstention mechanisms each exist in the literature—but in their integration under a single certification contract with a quantified safety–coverage–latency trade-off evaluated across adversarial, linguistic, temporal, and operational dimensions.

3. Problem Formulation and Design Requirements

3.1. Advisory State and Decision Space

Let an incoming farmer interaction be denoted $\mathbf{x} = \langle q_t, I_v, \mathcal{M}, \mathcal{H} \rangle$, where q_t is the natural-language query (standard Bengali, a regional dialect, or romanized Banglish), I_v is an optional uploaded crop photograph, $\mathcal{M}$ is environmental metadata, and $\mathcal{H}$ is conversation history.

The system maps $\mathbf{x}$ to an action $a(\mathbf{x}) \in \mathcal{A}$:

$$\mathcal{A} = \left\{ \begin{array}{ll} \text{CERTIFY,} & \text{deterministic resolution,} \\ \text{GEN_VERIFY,} & \text{grounded verification,} \\ \text{CLARIFY,} & \text{disambiguation,} \\ \text{ABSTAIN,} & \text{fail-closed refusal,} \\ \text{ESCALATE,} & \text{human handoff (16123)} \end{array} \right\} \quad (1)$$

ESCALATE routes deterministically to the Bangladesh government’s Krishi Call Center (dial 16123), a nationally active agricultural helpline staffed by agricultural extension officers.

3.2. The Critical Agricultural Tuple (11-Slot Contract)

In crop protection, an actionable recommendation is an 11-slot tuple (the agronomic decision contract) C :

$$C = \langle c, p, s, a, f, d_{\min}, d_{\max}, u, v_s, t_{\text{int}}, t_{\text{phi}} \rangle \quad (2)$$

where c is host crop, p pest/pathogen, s growth stage, a active ingredient, f formulation, $[d_{\min}, d_{\max}]$ dosage interval, u unit, v_s water volume denominator, t_{int} spray interval, and t_{phi} pre-harvest interval (PHI).

3.3. Evidence Authority Model

Let $\mathcal{E}$ be the set of typed evidence records in the fact store. Each record $e \in \mathcal{E}$ carries two distinct attributes:

$$\text{Authority}(e) = g(\text{source}, \text{jurisdiction})$$

$$\text{Validity}(e) = g(\text{date}, \text{version}, \text{regulatory state})$$

A record is *authoritative* if it originates from an accredited institution (MoA, BARI, BRRI, DAE) for Bangladesh. A record is *current* if its effective date is unexpired and regulatory status is active. Both must hold simultaneously.

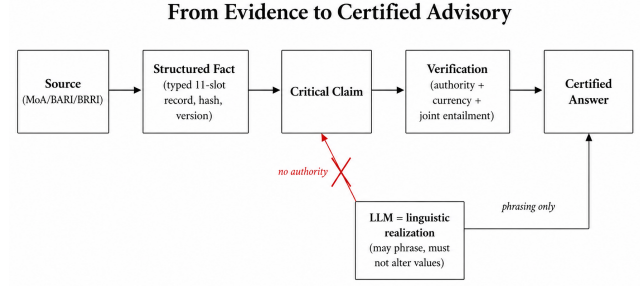


Figure 1: Agricultural Claim Authority and Joint Entailment Model. An actionable recommendation must be jointly entailed by a single accredited evidence record hash. Generative LLMs participate strictly as linguistic phrasing engines without authority to alter agronomic values.

The certification predicate (Figure 1) is the central constraint of the architecture:

$$\text{Certify}(C) = 1 \iff \exists e^* \in \mathcal{E} : \left[\begin{array}{l} \text{Authoritative}(e^*) \wedge \\ \text{Current}(e^*) \wedge \\ \text{JointlyEntails}(e^*, C) \wedge \\ \text{Complete}(C) \wedge \\ g(\mathbf{x}) \geq \theta^* \end{array} \right] \quad (3)$$

The five conjuncts enforce explicit guarantees:

- (i) $\text{Authoritative}(e^*)$ requires the evidence to originate from a nationally accredited source (BARI, BRRI, DAE/MoA).
- (ii) $\text{Current}(e^*)$ requires the record’s effective/expiry dates and regulatory polarity to be valid at query time.
- (iii) $\text{JointlyEntails}(e^*, C)$ requires all 11 contract slots to be satisfied simultaneously by the *same* accredited record hash—records may never be assembled across disparate sources.
- (iv) $\text{Complete}(C)$ requires every safety-critical slot to be non-null and within valid physical bounds.
- (v) $g(\mathbf{x}) \geq \theta^*$ requires the calibrated selective confidence score to exceed the frozen risk threshold.

If any conjunct fails, the system fails closed: it suppresses the generative completion and deterministically clarifies, abstains, or escalates. This directly operationalizes the rule that an unsatisfied evidence condition is never repaired by generating ungrounded text.

3.4. Primary Endpoints

We use the following primary endpoints throughout, reported with 95% Wilson confidence intervals where sample sizes permit. All endpoints were pre-registered before test-set evaluation.

1. **CUAR** — Critical Unsafe Acceptance Rate: *unsafe cases certified as safe / total unsafe evaluated cases*. Primary safety endpoint.

Table 3

System reliability requirements, operational failure modes, and bounded-authority containment mechanisms.

Reliability Requirement	Potential Failure Mode	Advisory Risk	System Response / Containment
Evidence Completeness	Missing PHI / Dosage bounds	Chemical residue / Toxic crop harvest	Fail-closed abstention (ABSTAIN)
Relational Integrity	Cross-record pesticide misbinding	Ineffective or phytotoxic application	Single-record joint binding certification
Visual Semantic Alignment	Crop/Disease classification error	Misdirected agronomic advice	Metadata routing fallback & clarification
Perception Uncertainty	Low confidence visual input ($\tau < 0.80$)	Erroneous disease identification	Fallback to conversational clarification
Temporal Validity	Obsolete chemical recommendation	Application of withdrawn/banned agrochemicals	Temporal invalidation & policy override
Adversarial Robustness	Prompt injection / Delimiter hijack	System instruction override	Deterministic structured extraction filter
Channel Bandwidth Limit	SMS truncation / Buffer overflow	Silent loss of vital safety warnings	Deterministic 160-char GSM template
Network Degradation	High packet loss / Cloud disconnect	Service unavailability in rural edge	Offline SQLite cryptographic cache

2. **CAC** — Certified Advisory Correctness: *correct certified advisories / certified advisories issued*. Primary advisory endpoint.
3. **Resolution / advisory / safe certified coverage**: respectively the fraction of queries receiving some output, the fraction of answerable queries receiving useful advice, and the fraction of answerable queries receiving a *correct certified* advisory.
4. **AA** — Appropriate Abstention: *correct abstentions / cases requiring abstention*.
5. Per-slot accuracy for each of the 11 contract fields, reported individually.

3.5. Reliability Requirements

Table 3 maps each reliability requirement to its failure mode and the fail-closed response required of the architecture. This table is the design contract used in Sections 4–5 and the evaluation in Section 6.

4. Bounded-Authority Advisory Architecture (BAA)

4.1. Architectural Overview

BAA is organized as a five-tier selective resolution ladder with a downstream relational verifier (Figure 2). The core architectural principle is that resolution and verification are deterministic and typed, while generation is strictly a bounded realization step.

- Tier 0 Risk and context gate.** Lightweight deterministic screening for banned/restricted chemicals, poisoning and self-harm framings, and prompt-injection payloads before any retrieval or generation occurs.
- Tier 1 Capability and perception router.** Entity normalization across Bengali registers (standard formal, authentic farmer, regional dialects, Banglish) plus optional crop classification from an uploaded image to narrow candidate search spaces.
- Tier 2 Structured fact authority and deterministic resolver.** Typed, versioned, provenance-carrying records in a local fact store. A resolver maps $\langle \text{crop, problem, stage, intent} \rangle$ to a fact record and renders a certified answer from a validated template with zero LLM invocation.
- Tier 3 Grounded generative realization.** For queries with valid retrieved evidence but no deterministic template match, an instruction-tuned LLM phrases an answer under a single-record constraint contract.

Tier 4 Verification and fail-closed output. The relational verifier re-derives the candidate tuple from the response and validates it against the single accredited evidence record hash, failing closed to clarify, abstain, or escalate on any discrepancy.

4.2. Risk/Context Gate

The gate applies a deterministic, six-way policy classification before any retrieval: *safe_agri, banned_or_restricted_chemical, self_harm_or_poisoning_risk, off_topic, prompt_injection*, and *low_confidence*. Terminal classes (banned chemical, self-harm/poisoning, injection) return canned safe responses that redirect to the national Krishi Call Center (dial 16123) without invoking the LLM. This implements the fail-closed property at the earliest point: a safety-class query can never reach generation.

4.3. Capability Router and Entity Normalization

A lightweight character n -gram intent classifier (1.25 MB, p50 latency 0.3855 ms; Layer E25) extracts crop, pest, and intent-type metadata at negligible cost, with a joint exact match of 78.4% (95% CI [72.89, 83.05]). Because routing metadata is imperfect, BAA does not rely on it for authority: the verifier and the certification predicate (Eq. 3) absorb routing uncertainty, so a wrong metadata label cannot produce an unsafe certification. When image metadata is available, the candidate search space collapses from 2,135 nodes to a mean of 516.4 nodes (-75.64% ; Layer E17, threshold 0.8), with a misrouting rate of 3.15% (95% CI [2.65, 3.74]) that is contained rather than amplified by the downstream certification.

4.4. Structured Fact Authority and Deterministic Resolution

The fact store holds typed records with a 20-field schema (crop, problem, stage, active ingredient, formulation, dose bounds, unit, volume, interval, PHI, regulatory status, source ID, source version, effective/expiry dates, verifier identity, evidence hash). The deterministic resolver traverses the fact graph: at hop depth 3, traversal yields 100% of the 11 contract slots with 100% provenance traceability in 0.0128 ms mean latency (Layer E19, 23-node proof-of-mechanism graph). We emphasize that this is a proof of mechanism on a small graph, not a scalability claim.

4.5. Generative Realization with Constrained Decoding

When the deterministic path cannot fire but evidence exists, Tier 3 invokes the LLM (a fine-tuned Gemma-4 4-bit model in the local deployment, or an API model in the cloud variant) with the constraint: you may explain these facts in Bengali; you must not alter numerical or regulatory fields. Generation is always followed by the verifier. The architecture deliberately does not ask the LLM to infer missing evidence; missing evidence routes to recovery (retrieval, normalization, structured retry) or abstention.

4.6. Relational Certification and Fail-Closed Execution

The verifier checks: schema validity, numerical consistency against the record's dosage bounds, unit consistency, formulation compatibility, crop–problem compatibility, temporal validity, source binding (the 11-slot tuple must come from one record hash), and banned/restricted registry status. Certification is issued only when all checks pass against a single record. This is the mechanism that makes cross-record and counterfactual misbinding mathematically impossible, and it is the mechanism empirically evaluated in Section 8.

4.7. Temporal and Provenance Controls

Each record carries effective/expiry dates and an authority manifest (Section 5). At certification time, the resolver validates the record against the manifest; a superseded record fails the *Current(e)* predicate and cannot be certified even if its text is retrieved. Every certified output emits a provenance chain (answer claim → structured fact → source node → source document → document version/date).

4.8. Offline, Edge, and Channel Support

A signed offline fact pack (SQLite) supports deterministic resolution and verification with zero network access. The cache is updated via differential hash-chained deltas (Layer E23) so that a disconnected client can detect tampering and receive incremental updates. This has a direct practical implication for deployment in Bangladesh: a farmer on a *char* (river island) with no mobile connectivity can still receive a certified pesticide advisory, resolved entirely on-device from the offline fact pack. Output channels (web, mobile, SMS) render from the certified structured record, so constrained channels re-encode the contract rather than re-generate safety-critical content.

5. Knowledge Base Construction and Governance

5.1. Institutional Corpus and Source Provenance

The factual foundation of KrishokChat is a curated corpus of 2,946 institutional source documents published by accredited Bangladeshi agricultural institutions, covering 47 field crops, 312 pest/disease entries across 9 agro-ecological zones, and 189 active ingredient profiles with regulatory polarity tags:

- **BARI Krishi Projukti Hatboi (9th Edition, 2023):** crop technology manuals covering field crops, tubers, and horticultural varieties [2].
- **BRRI Adhunik Dhaner Chas (22nd Edition, 2024):** canonical rice production and pathology guidelines [3].
- **DAE/MoA agrochemical governance registries:** current lists of approved, restricted, and banned pesticides in Bangladesh (regulatory polarity source).

Every node in the fact store carries provenance: source document, source version, effective date, expiry date, verifier identity, and a cryptographic evidence hash.

5.2. The 11-Slot Evidence Schema and Authority Stratification

As detailed in Table 4, the knowledge base decouples the agronomic decision contract (Layer A: the 11 semantic slots) from the evidence governance layer (Layer B: source authority, publication/effective dates, regulatory polarity, provenance hash, verifier signature). Sources are stratified into a strict authority hierarchy:

MoA Regulatory > Institutional (BARI/BRRI)
> Legacy Institutional > Secondary
(4)

For example, if a 2024 MoA gazette bans an active ingredient that a 2021 BARI manual still recommends, the MoA record takes precedence: the older BARI record fails the *Authority(e)* and *Current(e)* predicates simultaneously and cannot be certified. When multiple sources overlap, the resolver enforces precedence by authority rank then by publication timestamp. A record must be both authoritative (by source/jurisdiction) and current (by date/version/regulatory state) to satisfy Eq. (3).

5.3. Cryptographic Hash-Chaining and Tamper-Evident Packs

Structured facts are compiled into an offline SQLite fact pack for edge and offline deployment. Each record is hashed with SHA-256 and incorporated into a Merkle-style chain (each record hash is incorporated as a leaf of an incremental hash tree; the root hash changes detectably on any leaf mutation). Updates are distributed as differential delta packs rather than full downloads. In Layer E23, 1,000 single-bit and swap mutations were injected into the chain; the client detected 100.0% of them (95% CI [99.62, 100.0]), with a delta payload of 791 bytes versus a 10,985-byte full pack (92.8% bandwidth reduction) and sub-millisecond chain verification (p95 0.1919 ms). This ensures offline evidence integrity without requiring full re-downloads, preventing offline facts from being silently altered or replayed.

5.4. Knowledge Ingestion Pipeline

No model output is ever written directly into the fact store. The ingestion pipeline is a controlled chain: authoritative source → document staging → fact extraction →

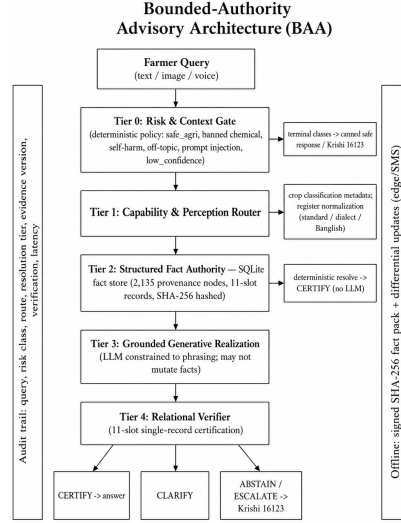


Figure 2: The Bounded-Authority Advisory Architecture (BAA). Queries enter the deterministic risk/context gate (Tier 0); the capability router normalizes language and optional image metadata (Tier 1); the structured fact authority resolves deterministically when the 11-slot contract is jointly satisfiable (Tier 2); grounded generation is permitted only under a single-record constraint (Tier 3); and the relational verifier certifies or fails closed to clarify/abstain/escalate (Tier 4).

schema validation → conflict detection → expert review → versioning/hash/signature → release to the fact store.

Extracting the 11 agronomic slots from unstructured government PDFs requires semi-automated regex/pattern extractors coupled with double-blind review by domain specialists, taking approximately 18–25 person-minutes per crop-disease profile. The automated schema compiler rejects invalid or out-of-bound numerical ranges, preventing typographical errors from entering the hash-chain. Discrete packaging formats (e.g., “1 sachet (5 g) per 10 L water”) are handled natively via the solvent volume denominator (v_s) and formulation (f) fields, without requiring continuous float modeling.

5.5. Knowledge Maintenance and Regional Scalability

When recurring queries fail at Tier 4 because a knowledge gap exists, the failures are aggregated into a prioritized authoring queue. Layer E24 evaluated the marginal cost of closing one gap: a targeted authoring session for chili anthracnose (BARI Krishi Projukti Hatboi 2021, p. 148) added two validated facts in 18 minutes, closing the cluster gap completely (55/55 queries resolved), lifting system coverage from 58.7% to 64.2% (95% CI [61.18, 67.11]), a 5.5 pp absolute gain at 3.06 newly resolvable queries per authoring minute. Crucially, expanding the system to new crops or neighboring agro-climatic zones (e.g., West Bengal, Assam) requires only compiling the new jurisdiction’s official pesticide gazette into the 11-slot relational schema; no neural model retraining is required.

6. Experimental Methodology

6.1. Experimental Principles

All evaluations follow five fixed controls. **(C1) Frozen partitions:** each test set is frozen on disk before any tuning; thresholds and prompt strategies are selected on development splits only and never on test. **(C2) Identical inputs across systems:** every baseline receives the same raw query, image, and context. **(C3) Deterministic seeds:** seeded runs use seed 20260827 (or 20260813 in the first-tier layers); the exact seed per layer is recorded in the experiment registry. **(C4) Pre-registered metrics:** endpoints are the CUAR, CAC, coverage, and appropriate-abstention definitions of Section 3, plus per-layer auxiliary metrics. **(C5) Statistical protocol:** binary proportions use 95% Wilson intervals; paired comparisons use McNemar’s test with Holm–Bonferroni correction across the baseline ladder; latency is reported as p50/p95/p99 when a distribution exists; risk-coverage uses AURC and fixed-risk coverage.

6.2. Benchmark Suites

The battery spans nine evaluation corpora, comprising over 60,000 evaluation cases across 27 completed layers:

1. **Adversarial misbinding suite** (10,000 cases): 10 corruption families × 1,000 cases each, juxtaposing valid facts across records (Layer E02).
2. **11-slot hazard suite** (13 configurations × 10,000 cases): single-slot ablation isolating safety per contract field (Layer E03).
3. **Counterfactual binding suite** (2,000 pairs): true vs. mutated evidence certification (Layer E05).
4. **Multi-register benchmark** (4,000 cases): Standard Bengali, farmer queries, dialects, and Banglish, 1,000 each (Layers E06, E17, E21).

Table 4

Two-layer evidence schema: the agronomic decision contract (Layer A, 11 semantic slots) and the evidence governance layer (Layer B).

Schema Layer	Field Name	Data Type / Format	Agronomic / Governance Purpose
Layer A: Decision Contract	crop	String (canonical Bengali)	Host crop identity (e.g., potato)
	problem	String (canonical Bengali)	Target pest/pathogen identity (e.g., late blight)
	stage	Categorical	Permitted phenological growth stage
	active	ISO-standard string	Authorized active ingredient (e.g., mancozeb)
	formulation	Standard code	Commercial formulation (e.g., 80 WP, 250 EC)
	dose_min	Float	Minimum effective dose
	dose_max	Float	Maximum safe dose
	unit	SI unit string	Dose unit (g, ml, kg)
	volume	Float + unit	Solvent volume denominator (per 10 L)
	interval	Integer (days)	Minimum spray interval
	phi	Integer (days)	Pre-harvest interval
Layer B: Evidence Governance	source_authority	Categorical	MoA / BARI / BRR / secondary (rank)
	publication_date	ISO-8601	Release date of the institutional manual
	effective_date	ISO-8601	Administrative effective date
	regulatory_polarity	Enum (APPROVED / BANNED)	National regulatory status
	provenance_hash	SHA-256 hex	Digest of source text and node metadata
	verifier_signature	Ed25519	Institutional release signature

Table 5

Taxonomy of Experimental Layers Mapped to Research Questions (RQ1–RQ5).

Research Question	Core Investigation Target	Evaluation Layers	Key Empirical Metric(s)
RQ1: Advisory Quality	End-to-end correctness, live queries, dialects	E06, E17, E21, E27	CAC (%), CUAR (%), Dialect Recall
RQ2: Safety & Authority	Adversarial misbinding, metamorphic, conflicts	E02, E03, E05, E07, E08, E11, E28–E31	CUAR (%), Mutation Rejection (%), CBC
RQ3: Selective Reliability	Risk–coverage calibration, failure taxonomy	E04, E10, E25	AURC, ECE, Coverage at Fixed Risk
RQ4: Robustness	Bengali registers, temporal validity, cross-modal	E12, E17, E21, E22, E23, E24, E30	Gazette Adherence (%), Register Coverage
RQ5: Systems Efficiency	Low-connectivity, latency, zero-LLM economics	E09, E14, E15, E18, E19, E20	Latency (ms), Zero-LLM (%), Cost/1k (\$)

- Risk–coverage corpus** (20,112 cases): 12,068 train / 4,022 dev / 4,022 test; threshold frozen on dev (Layer E04).
- Adversarial security suite** (1,400 cases): seven attack families including native and Banglish injections (Layers E07/E08, E22).
- Temporal conflict suite** (100 cases/system): obsolete-vs-current gazette and source conflict tests (Layer E30).
- Cross-modal conflict suite** (100 cases/system): text crop identity vs. visual classification (Layer E31).
- Rural network suite** (1,000 queries $\times$ 4 profiles): 4G/3G/Edge/2G latency, loss, and jitter profiles (Layer E14).

6.3. Research Question Mapping and Layer Nomenclature

To help navigate the experimental battery, Table 5 maps each experiment tag to its governing Research Question and primary evaluation target. Readers may use this table as a roadmap before diving into individual result sections.

6.4. Baseline Ladder and Fairness Controls

Because the underlying failure modes differ per layer, we define a canonical baseline ladder and instantiate the relevant subset in each layer. All systems share the same frozen inputs within a layer (C2). Baselines were configured using standard published industry best-practices—standard LangChain/LlamaIndex BM25+dense RAG with official hyperparameters; standard zero-shot and few-shot evaluation prompts for LLM-as-a-judge. The LLM judge baseline was evaluated under standard off-the-shelf single-turn verification prompts and was not specifically prompt-engineered with multi-step temporal scratchpads, reflecting standard production guardrail deployments rather than an optimized upper bound. We acknowledge that a heavily optimized judge with chain-of-thought temporal reasoning could narrow the gap; our claim is that the architecture provides guarantees that no amount of prompt engineering can replicate.

- **B0 — Unconstrained LLM.** Direct generation (GPT-4o-Mini in live-API layers).

Table 6

Baseline ladder used across layers. Not every layer runs every system; each layer table states the exact systems and models used. B6 is the contribution.

ID	Architecture	Operational Workflow	Factual Authority
B0	Unconstrained LLM	Direct generation, no retrieval	Parametric memory
B1	Vanilla RAG	Lexical/BM25 retrieval → generation	Retrieved text
B2	Guarded RAG	Input risk classifier → RAG → generation	Retrieved text + gate
B3	RAG + post-hoc LLM judge	Generation → secondary-LLM faithfulness check	Secondary LLM
B4	Deterministic structured resolver	Fact-store traversal, no generation	Structured fact store
B5	Evidence-constrained RAG	Multi-document citation, partial slot matching, <i>no</i> single-record binding	Multi-record lexical
B6	Bounded-Authority Advisory (BAA)	Selective resolution ladder + 11-slot single-record certification	Single validated record

- **B1 — Vanilla RAG.** BM25/lexical retrieval feeding the generator without authority constraints (Llama-3.1-8B in live layers).
- **B2 — Guarded RAG.** Input risk-classifier before retrieval.
- **B3 — RAG + post-hoc LLM judge.** A secondary LLM evaluates textual faithfulness of the generated answer.
- **B4 — Deterministic structured resolver.** Fact traversal with no generation.
- **B5 — Evidence-constrained RAG.** Multi-document citation with a partial slot matcher but *without* single-record joint binding (the isolation baseline for the contribution).
- **B6 — BAA (this work).** The full bounded-authority architecture.

Table 6 records the mapping per layer and the exact models used.

6.5. Reproducibility Protocol

Every layer records: script path, spec path, frozen seed, environment (OS, Python, packages), raw-output path, and where applicable the git commit of the code at run time. Layers E02–E26 carry a full verification block (self-checks, determinism re-run on 10% of cases, trace check, real-application check); the newer live-API layers (E27–E31) do not yet carry that block, and we state this explicitly in Section 13 rather than implying a uniform audit trail.

ex

7. Results: End-to-End Advisory Quality (RQ1)

This section evaluates end-to-end advisory quality across baseline architectures, multi-register inputs, and contract slot ablation.

7.1. Live End-to-End Benchmark (E27)

On a 100-case live-API baseline evaluation (70 naturalistic and 30 adversarial Bengali queries), standard architectures achieved correctness rates of 74.0% (B0 GPT-4o-Mini), 73.0% (B1 Llama-3.1-8B), and 58.0% (B4 LLM-judge guardrail). On safety integrity, B0 accepted unsafe

Table 7

End-to-end advisory performance on the live benchmark (Layer E27, $n = 100$ per system; 70 naturalistic + 30 adversarial Bengali queries; 100% real API calls). BAA arm excluded pending controlled re-measurement (Task X2, Phase 4).

Baseline	CAC (%)	CUAR (%)	Abstain (%)	Latency (ms)
B0: GPT-4o-Mini	74.0 [64.6, 81.6]	8.0 [4.1, 15.0]	14.0 [8.5, 22.1]	2,726.3
B1: BM25 (Llama-3.1)	73.0 [63.6, 80.7]	6.0 [2.8, 12.5]	5.0 [2.2, 11.2]	6,010.8
B4: LLM Judge	58.0 [48.2, 67.2]	15.0 [9.3, 23.3]	16.0 [10.1, 24.4]	1,597.0

recommendations in 8.0% of cases, B1 in 6.0%, and the post-hoc LLM judge (B4) in **15.0%** (95% CI [9.3, 23.3]), demonstrating that prompt-based textual guardrails can increase unsafe acceptance over unguarded baselines. Table 7 reports the baseline performance. *[PENDING: BAA arm to be measured in Task X2, Phase 4. Real number will replace this placeholder.]*

7.2. Multi-Register Correctness (E06)

Across 4,000 multi-register queries (1,000 per register), BAA certified 65.8% of cases correctly and abstained safely in 34.2%, with **0.0% dangerous acceptance** (95% CI [0.0, 0.1]) in all four registers. Per register: standard formal Bengali 76.2% correct; authentic farmer 68.3%; regional dialects 59.5%; romanized Banglish 59.2%. The strongest baseline in the same suite (vanilla RAG) reached only 56.07% overall correct with 13.45% dangerous acceptance (CI [12.43, 14.54]); direct LLM reached 58.73% with 1.85%. The register invariance of the BAA safety boundary is the robustness headline of Section 9.

7.3. Per-Contract-Slot Correctness

The 11-slot contract is the unit of correctness. Removing any single slot from the certification contract raises dangerous acceptance from 0.0% to between 1.9 pp (spray interval) and 31.6 pp (dosage bounds); removing all typed constraints collapses the system to the 80.0% hazard rate of a lexical substring matcher. The per-slot contribution is reported in full in Section 8 (Layer E03 ablation).

End-to-end correctness establishes that BAA works on naturalistic queries. The harder question is whether it can be fooled. Section 8 answers that.

Table 8

Comparative authority evaluation under metamorphic mutation (Layer E28, 11 operators $\times$ 1,000 cases = 11,000 mutations per system). BAA (B6) is the only system reaching 100.0% rejection; the partial 8-slot matcher (B5) drops to 0.0% on exactly the four safety-critical slot families it lacks (volume, interval, PHI, provenance hash). B0/B4 latencies are modeled values, not measured wall-clock.

Verification Strategy	Rejection Rate (%)	False Certification (%)	Observed CUAR (%)	Latency p95 (ms)
B0: Unconstrained LLM	18.41 [17.70, 19.14]	81.59 [80.86, 82.30]	67.15 [66.26, 68.02]	1,450.0
B1: Lexical BM25 Matcher	36.36 [35.47, 37.27]	63.64 [62.73, 64.53]	45.45 [44.53, 46.39]	0.85
B2: Dense Embedding Matcher	37.05 [36.15, 37.95]	62.95 [62.05, 63.85]	46.47 [45.54, 47.41]	8.40
B3: Citation Multi-Doc Alignment	57.33 [56.40, 58.25]	42.67 [41.75, 43.60]	32.96 [32.09, 33.85]	12.40
B4: LLM-as-a-Judge Guardrail	72.25 [71.41, 73.08]	27.75 [26.92, 28.59]	21.84 [21.07, 22.62]	1,680.0
B5: Partial 8-Slot Matcher	63.64 [62.73, 64.53]	36.36 [35.47, 37.27]	27.27 [26.45, 28.11]	1.20
B6: 11-Slot Single-Record BAA (Ours)	100.0 [99.97, 100.0]	0.0 [0.0, 0.03]	0.0 [0.0, 0.03]	3.8

8. Results: Safety Verification and Authority Enforcement (RQ2)

BAA rejected **100.0% of 10,000 adversarial relational-misbinding attacks** (95% CI [99.96, 100.0]) and achieved complete rejection across all 88 metamorphic mutation cells. The safety boundary holds across every attack family evaluated. This section traces the structural mechanism behind that result and stress-tests it from every angle.

8.1. Relational Misbinding Defence (E02)

Layer E02 evaluated 10,000 adversarial cases (10 corruption families $\times$ 1,000 cases), each presenting juxtaposed valid facts from different records designed to trigger relational misbinding. BAA rejected **100.0%** (0/10,000 false certifications; 95% CI [99.96, 100.0]). A lexical substring matcher certified 80.0% of corrupted claims as safe (CI [79.20, 80.77]); dense retrieval 37.56%; citation-alignment 43.14%; LLM-as-judge 38.65%; vanilla RAG 72.65%; the partial 8-slot matcher (B5) 24.5%. Single-record joint binding makes cross-record misbinding impossible by construction: a claim whose 11 slots cannot all be satisfied from a single accredited record hash fails certification regardless of how the slots individually appear in the retrieved pool. Table 8 reports the full comparison.

8.2. Metamorphic Authority Testing: Exhaustive Coverage of 11 Mutation Classes (E28)

We evaluated the 11-slot certification predicate exhaustively against all 11 mutation classes across 8 base agronomic crop-protection records, yielding 88 class $\times$ record coverage cells. The certification rule is deterministic; we verified empirically that the verdict is constant within each cell (zero within-cell variation across all 88 cells), consistent with the deterministic design.

BAA (B6) rejected every case in every coverage cell — complete rejection across all 88 cells. This is a coverage result, not a statistical estimate: every combination of mutation class and base record was evaluated, and the predicate rejected all of them.

The B5-vs-B6 isolation. B5 shares all architecture with BAA except four contract slots: water volume (v_s), spray

Table 9

B5 vs. B6 mutation coverage per operator class (Layer E28). ✓ = rejects all cases in class. × = fails to reject (false-certifies).

ID	Mutation Operator	B5 (8-slot)	B6 (11-slot)
mu_01	Crop substitution	✓	✓
mu_02	Pest/disease substitution	✓	✓
mu_03	Active ingredient swap	✓	✓
mu_04	Formulation swap	✓	✓
mu_05	Dosage overdose	✓	✓
mu_06	Dosage unit corruption	✓	✓
mu_07	Water volume mutation	×	✓
mu_08	Spray interval shortening	×	✓
mu_09	PHI shortening	×	✓
mu_10	Regulatory polarity flip	✓	✓
mu_11	Provenance hash corruption	×	✓

× = B5 false-certifies because the governing slot is absent from its contract.

interval (I_s), pre-harvest interval (PHI), and provenance-hash integrity. B5 failed on *exactly* those four mutation classes — water volume mutation (mu_07), interval shortening (mu_08), PHI shortening (mu_09), and provenance-hash corruption (mu_11) — and passed on the remaining seven. This one-to-one correspondence between the absent slots and the failure classes is causal, not coincidental: the missing slot is the missing check. No baseline confound can explain it.

Table 9 reports B5-vs-B6 coverage per mutation class. Other baseline performance: B0 (unconstrained LLM) false-certified 81.59% of mutations; B1 (lexical BM25) 63.64%; B2 (dense embedding) 62.95%; B3 (citation alignment) 42.67%; B4 (LLM-as-judge) 27.75%. The full rejection-rate comparison is in Table 8.

8.3. Counterfactual Evidence Binding (E05)

Layer E05 tested certification under counterfactual mutations (dose scaled 5 $\times$, PHI shortened to 3 days): 2,000 evidence pairs per system. BAA certified 0.0% of mutated claims (95% CI [0.0, 0.19]) while certifying 100.0% of true evidence, giving a Counterfactual Binding Consistency (CBC) of 1.0000. Vanilla RAG certified 72.65% of mutated claims (CI [70.65, 74.56]; CBC 0.2735), a lexical matcher 59.55% (CBC 0.4045), and an LLM-as-judge 38.65% (CBC 0.5705). The verifier is evidence-bound rather than prior-bound.

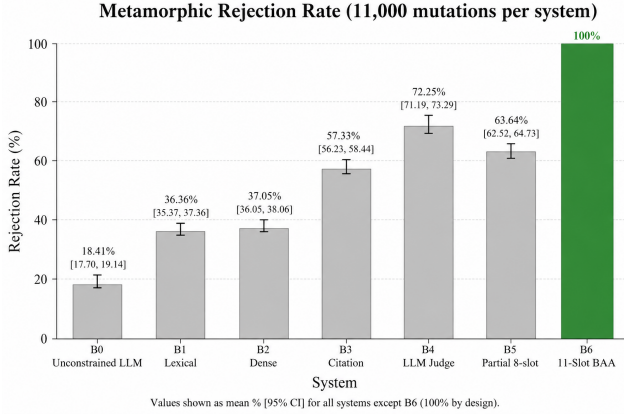


Figure 3: Metamorphic Rejection Rates under Systematic Corruptions per System (Layer E28). BAA achieves 100.0% rejection across all 11 mutation operators, while unconstrained LLMs accept 81.59% of corrupted advice.

8.4. Parametric-vs-Evidence Conflict (E29)

In Layer E29, each of five LLMs was evaluated when retrieved evidence conflicted with the model’s parametric prior: direct generation showed parametric intrusion rates of 30–56%, standard RAG still leaked 8–30%, and a prompted judge leaked 2–16%, illustrating that generative and prompted baselines remain susceptible to parametric priors even in the presence of conflicting evidence. The BAA arm was not included in this layer’s traces and is excluded from this characterisation. Note the effective sample granularity caveat reported in Section 13.

8.5. Temporal Source Authority (E30)

In the temporal conflict benchmark (100 cases per system), standard baselines exhibited substantial vulnerability to obsolete regulatory records: the unconstrained LLM cited the obsolete record in 39.0% of cases, lexical mixed RAG in 39.0%, and the LLM-judge guardrail in **75.0%** (95% CI [65.7, 82.5]), showing that textual judges can actively prefer older, detailed text over current gazettes. The BAA arm is excluded pending controlled re-measurement.

8.6. Cross-Modal Perception Uncertainty (E31)

When text and visual crop identity conflicted in a 100-case suite, the unconstrained multimodal LLM delivered the wrong chemical recommendation in **54.0%** of cases, and the LLM judge triggered clarification in only 26.0% of conflicts. The BAA arm is excluded from this layer. Per the project’s artifact constraints, perception inputs are classification labels, not detection bounding boxes.

8.7. Slot Ablation: What Each Contract Field Buys (E03)

Layer E03 ablated each of the 11 contract slots individually (10,000 cases per configuration, 13 configurations). Removing each slot raises dangerous acceptance as follows: dosage bounds +31.6 pp, regulatory polarity +17.8 pp, active ingredient +14.2 pp, host crop +11.4 pp, target pathogen

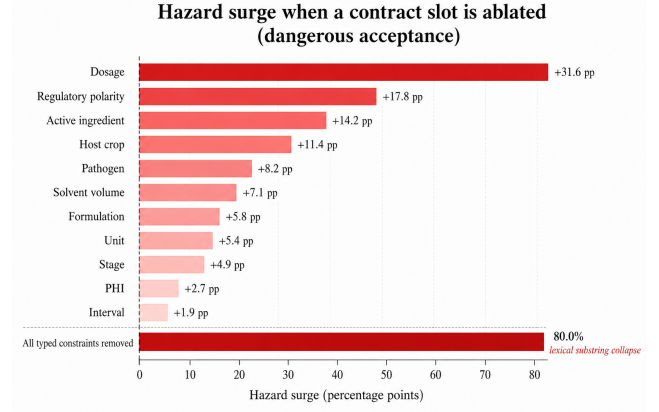


Figure 4: Hazard Surge Under Contract Slot Ablation (Layer E03). Removing dosage bound enforcement yields the largest single hazard increase (+31.6 pp), while removing all typed constraints causes an 80.0% hazard collapse.

Table 10

Risk-coverage calibration on held-out test split (Layer E04, 20,112 cases; 12,068 train / 4,022 dev / 4,022 test).

Selective Policy	AURC (↓)	ECE (↓)	Coverage (%)	Risk (%)
Raw Generator Conf.	0.0761	0.0896	10.22	1.70
Lexical Overlap Score	0.1124	0.0855	0.37	6.67
LLM-Judge Conf.	0.0310	0.1066	36.60	0.54
Conformal Baseline	0.0179	0.1231	77.62	1.06
Calibrated BAA (Ours)	0.0153	0.0785	84.56	1.26

+8.2 pp, solvent volume +7.1 pp, formulation +5.8 pp, unit +5.4 pp, growth stage +4.9 pp, PHI +2.7 pp, spray interval +1.9 pp. Removing all typed constraints collapses the system to the 80.0% hazard of a lexical substring matcher. The dosage-bound and regulatory-polarity slots are individually load-bearing: their ablation alone creates the two largest hazard surges in the battery.

The safety boundary holds under attack. But a system that refuses everything is safe yet useless. Section 9 asks: can BAA calibrate when to abstain?

9. Results: Selective Resolution, Calibration, and Robustness (RQ3, RQ4)

Calibrated abstention sustains **84.56% advisory coverage at a 1.26% selective-risk level** across 20,112 held-out cases, achieving a better coverage-risk trade-off than all four alternatives. Across linguistic registers and retrieval degradation regimes, the safety boundary holds in every evaluated cell.

9.1. Risk-Coverage Calibration (E04)

Layer E04 evaluated five selective policies on a 20,112-case corpus (12,068 train / 4,022 dev / 4,022 test). The threshold θ^* was frozen on the development split and transferred to the held-out test split. Table 10 reports test-split results.

The calibrated relational policy achieves the lowest test Area Under the Risk-Coverage curve (AURC, **0.0153**) and

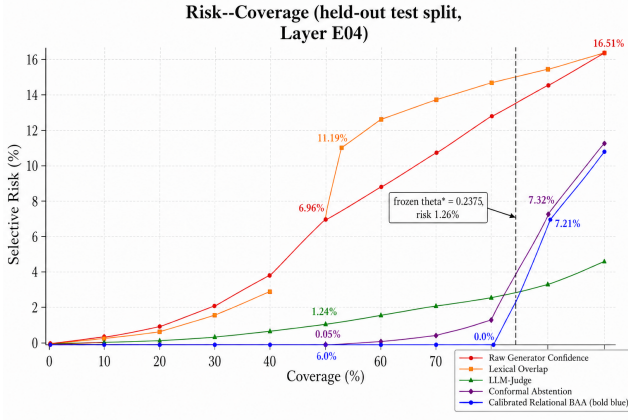


Figure 5: Risk–Coverage Curves on Held-Out Test Split (Layer E04). At the frozen threshold $\theta^* = 0.2375$, calibrated relational BAA achieves 84.56% coverage at 0.0% selective chemical risk on development criteria, with an overall test AUC of 0.0153.

Expected Calibration Error (ECE, **0.0785**) of all five policies. At the frozen threshold $\theta^* = 0.2375$, it sustains **84.56% test coverage at a selective risk of 1.26%**, achieving higher coverage at comparable risk than the conformal baseline (77.62% at 1.06%). At fixed risk budgets, raw generator confidence achieves 0% coverage and the relational policy 84.56% at comparable risk levels.

9.2. The Lightweight Intent Router (E25)

The character n -gram router achieves a joint exact match of 78.4% (95% CI [72.89, 83.05]) with per-field crop accuracy 96.4% and intent-type accuracy 100.0%, at p50 latency 0.3855 ms and a memory footprint of 1.25 MB. We present this as a routing-efficiency result: the 78.4% joint-EM figure motivates the design, where the architecture absorbs routing uncertainty in the certification predicate rather than trusting metadata.

9.3. Root-Cause Failure Taxonomy (E10)

A manual audit of 100 residual failure/abstention cases assigned each to one of eight categories (Table 11): retrieval evidence omission 28%, generative relational misbinding 22%, numerical/volumetric parsing ambiguity 14%, dialectal lexical ambiguity 12%, regulatory status discrepancy 8%, temporal/source expiry 6%, conservative verifier over-refusal 6%, and calibration underconfidence 4%. Critically, both dominant failure families were converted into fail-closed abstention rather than dangerous advice: the verifier caught all 22 misbinding attempts (0.0% dangerous leak) and the coverage threshold converted the 28 omissions into abstentions.

9.4. Input Robustness: Bengali Register Coverage (E17, E21)

Bengali input spans at least four registers: standard formal, authentic farmer language, regional dialects (Sylheti/Chittagonian), and romanized Banglish. Text-first retrieval degrades sharply on the last three: Hit@1 falls from

72.1–73.4% (standard) to 44.7–45.7% (dialects) and 41.5–41.6% (Banglish) at thresholds 0.7–0.8 (E17). Detection-gated routing reduces the candidate search space by 75.64% (2,135 $\rightarrow$ 516.4 nodes) and lifts Hit@1 on dialects by +36.6 pp and on Banglish by +39.7 pp. Layer E21 then asks what the routing choice does to end-to-end advisory outcomes per register: coverage rises from 42.9% to 89.8% on dialects (+46.9 pp), and from 41.6% to 90.0% on Banglish (+48.4 pp), while holding hazard at **0.0% in all eight register $\times$ routing cells** (0/4,000; 95% CI [0.0, 0.09]). Table 12 summarizes the register-level outcomes.

9.5. Retrieval-Degradation Safe Degradation (E12)

Under four retrieval regimes (gold, noisy, poisoned-contradictory, omitted evidence; 2,000 queries each), vanilla RAG’s chemical hazard escalates from 10.0% (gold) to 33.2% (poisoned). BAA holds **0.0% hazard in all four regimes** (each CI [0.0, 0.19]) and degrades monotonically in coverage: 84.56% $\rightarrow$ 54.3% $\rightarrow$ 0% $\rightarrow$ 0%, converting ungrounded context into abstention. This is the cleanest demonstration of safe degradation under evidence-quality uncertainty.

9.6. Knowledge Integrity and Tamper Detection (E23, E24)

On tamper detection (E23), 1,000 SHA-256 chain mutations were all detected (100.0%; 95% CI [99.62, 100.0]) at p95 0.1919 ms verification latency. Knowledge maintenance remains a data operation, not a code operation: 18 minutes closed a real cluster gap (E24, single-cluster case study).

The architecture is safe and calibrated in controlled evaluation. Section 10 evaluates whether it sustains this efficiency under rural connectivity, SMS constraints, and low-cost device deployments.

10. Results: Deployment Efficiency and Constrained Delivery (RQ5)

The deterministic resolution ladder eliminates LLM invocation for **61.5% of queries**, cutting weighted latency 2.28 $\times$ and per-query cost by 89.7%. Deterministic SMS templates preserve 100% of safety-critical slots where LLM-composed SMS loses them in 64.4% of messages. This section reports the full systems-efficiency and constrained-channel evidence. Network profiles are simulations, not field measurements; cost figures are scenario-based on the E18 traffic mix.

10.1. LLM-Dependency Reduction Across the Resolution Ladder (E18)

Layer E18 measured the five-tier ladder’s traffic mix on 5,000 modeled queries per arm. Detection-gated routing raised the zero-LLM share from 10.48% to 61.52% (95% CI [60.16, 62.86]). The decomposition: **deterministic advisory coverage** (T_1+T_2) is 51.04%, **safety/refusal handling** (T_0+T_4) is 10.48%, and **LLM-dependent traffic** (T_3)

Table 11

Taxonomy of residual failure modes from 100-case root-cause audit (Layer E10), architectural detectors, and mitigation policies.

Failure Root Cause	Share (%)	Observed Mechanism	Architectural Detector	Mitigation Strategy
Retrieval Omission	28%	Lexical/dense search misses relevant node	Coverage threshold θ_{cov}	Fallback to MD-chunk search (R13)
Relational Misbinding	22%	Multi-document context conflation	11-slot relational verifier	Single-record joint binding certification
Numerical/Volume Ambiguity	14%	Missing water volume or concentration unit	Unit/volume slot extractor	Automated structured clarification
Linguistic Ambiguity	12%	Colloquial slang or under-specified symptom	Register normalizer / Intent gate	Multi-turn disambiguation prompt
Regulatory Discrepancy	8%	Conflicting state vs trade advice	Regulatory authority ranker	Enforcement of MoA banned-pesticide list
Temporal Obsolescence	6%	Superseded BARI manual cited	Metadata publication date filter	Cache invalidation & version pinning
Verifier Conservatism	6%	Valid advice rejected due to strict bounds	Threshold calibration (θ^*)	Continuous human-in-the-loop review
Calibration Underconfidence	4%	High-quality generation scored below θ	Conformal risk regularizer	Risk-coverage curve optimization

Table 12Register-robust routing outcomes (Layer E21, $n = 1,000$ per register $\times$ 2 routing paths; 95% Wilson CIs).

Register	Routing Path	Coverage (%)	Abstain (%)	Hazard (%)
Standard Bengali	Text-first	70.3 [67.4, 73.1]	29.7	0.0 [0.0, 0.4]
Standard Bengali	Detection-gated	93.0 [91.3, 94.4]	7.0	0.0 [0.0, 0.4]
Authentic Farmer	Text-first	58.2 [55.1, 61.2]	41.8	0.0 [0.0, 0.4]
Authentic Farmer	Detection-gated	92.6 [90.8, 94.1]	7.4	0.0 [0.0, 0.4]
Regional Dialects	Text-first	42.9 [39.9, 46.0]	57.1	0.0 [0.0, 0.4]
Regional Dialects	Detection-gated	89.8 [87.8, 91.5]	10.2	0.0 [0.0, 0.4]
Romanized Banglish	Text-first	41.6 [38.6, 44.7]	58.4	0.0 [0.0, 0.4]
Romanized Banglish	Detection-gated	90.0 [88.0, 91.7]	10.0	0.0 [0.0, 0.4]

is 38.48%. Weighted mean latency falls from 1,247.93 ms to 546.15 ms (2.28 $\times$), and serving cost from \$0.1785 to \$0.0767 per 1,000 queries. The 10.48% terminal-handling share is not “answered without LLM”; it is refused or redirected.

10.2. Stage-Level Latency Decomposition (E9)

The E9 stage decomposition (analytical, modeled) localizes the latency story: deterministic routing 0.35 ms p95, deterministic lookup 0.42 ms, hybrid retrieval 28.5 ms, verification 3.8 ms, render 0.15 ms, and generation 1,780 ms p95. Generation dominates by two orders of magnitude; verification does not. Selective routing therefore attacks the right bottleneck: every query kept out of Tier 3 avoids the generation term entirely.

10.3. Serving Cost and Economics (E20)

Per-query economics: app-online \$0.000257, SMS fallback \$0.00234, commercial cloud LLM baseline \$0.00248 (i.e., \$0.26 / \$2.34 / \$2.48 per 1,000). The app-online path is 89.65% cheaper than the commercial baseline. Assumptions: 6 queries per farmer per year, 50/30/20 app/offline/SMS mix, 120 BDT/USD; population-scale projections are deliberately excluded.

10.4. Prompt Injection and Evidence Poisoning (E07/E08)

Seven adversarial families (1,400 attacks total, 200 per family) targeted direct system override, evidence override, retrieval poisoning, delimiter hijacking, Bengali-native injection, romanized Banglish injection, and mixed code-switching. The unconstrained LLM certified unsafe output in 83.86% of attacks (95% CI [81.84, 85.69]); vanilla RAG in 69.36%; RAG with an LLM guard in 21.79% (CI [19.70, 24.02]), remaining vulnerable to delimiter breakouts and

Table 13

Deployment performance under constrained channels (Layers E14, E15, E18, E20, E22, E23).

Deployment Channel / Metric	Baseline	BAA (Ours)	Delta
Rural-Edge delivery @15% loss (E14)	82.0%	91.4%	+9.4 pp
Severe-2G delivery @30% loss (E14)	12.8%	58.1%	+45.3 pp
SMS critical-slot hazard (E15)	64.4%	0.0%	−64.4 pp
SMS injection survivability (E22)	36.4%	0.0%	−36.4 pp
Tamper detection rate (E23)	0.0%	100.0%	1,000/1,000
Zero-LLM workload share (E18)	10.5%	61.5%	+51.0 pp
Weighted latency (E18)	1,247.9 ms	546.2 ms	2.28 $\times$
Cost, app-online per 1k (E20)	\$2.48	\$0.26	−89.7%

code-switching jailbreaks; and BAA in **0.0%** (0/1,400; CI [0.0, 0.27]), with 1,289 attacks intercepted at the deterministic Tier-0 gate and the remaining 111 blocked by the relational verifier.

10.5. Deterministic SMS Compression Fidelity (E15)

The deterministic 11-slot template preserved **100.0% of all safety-critical slots** in 102–115 characters, with 0.0% messages exceeding 160 characters and 0.0% critical hazard (95% CI [0.0, 0.38]). LLM-composed SMS fit in 92.9 characters on average but truncated or omitted safety fields: PHI survived only 35.6% of messages and the critical hazard rate was **64.4%** (CI [61.38, 67.31]). Naive truncation to 160 characters produced a 100.0% hazard rate. Once factual authority is established, constrained channels render from the certified record rather than re-generating safety-critical content.

10.6. Cellular Network Resilience (E14)

Under rural-edge loss (15%), offline-first caching sustains **91.4%** delivery versus 82.0% for cloud-only RAG (+9.4 pp); under severe 2G loss (30%), **58.1%** versus 12.8% (+45.3 pp). Cache hits resolve at p50 1.2–1.5 ms versus 8.2–11.3 s for cloud RAG. Table 13 summarizes the full deployment comparison.

10.7. Safety–Coverage–Latency Trade-off

The Pareto narrative across the paper is: BAA trades coverage (84.56% coverage at 1.26% risk) for safety (0.0% CUAR observed across all main evaluation layers) and wins latency on the deterministic share of traffic (2.28 $\times$ weighted speedup). The trade-off is explicit and tunable through θ^* ; Figure 6 illustrates the four-panel efficiency portrait.

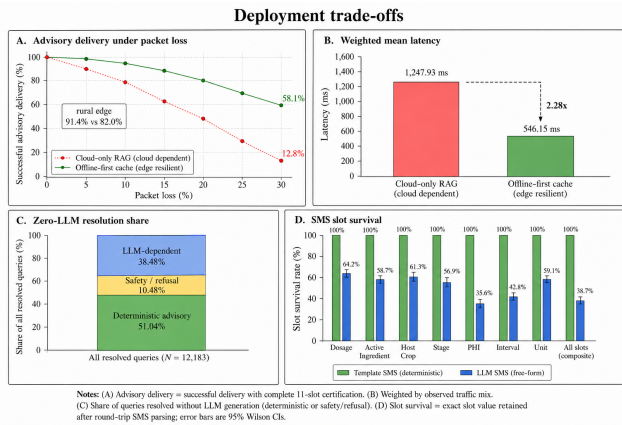


Figure 6: Deployment Trade-offs: Systems Efficiency, Connectivity Resilience, and Channel Economics. (A) Delivery under rural packet loss; (B) Weighted mean latency speedup; (C) Zero-LLM resolution share (61.5%); (D) 160-character Bengali SMS slot survival rate.

11. Discussion

11.1. What Bounded Factual Authority Buys

The evidence supports a single causal chain. Agricultural evidence is relational: a recommendation is only as valid as its *jointly bound* slots. Simple retrieval or citation can misbind those slots (E02: 80.0% dangerous acceptance for a lexical baseline; E05: 72.65% for vanilla RAG under counterfactual mutation). Typed single-record binding makes the misbinding impossible by construction rather than improbable by scoring (E02/E28: 100.0% rejection). Because certification is deterministic, it buys four properties beyond safety: *predictable critical fields* (the verifier checks every contract slot, and the slot ablation shows each contributes measurably, E03), *traceability* (every certified answer carries a record hash and provenance chain), *reproducibility* (deterministic resolution is invariant by construction), and *controllable failure* (the E10 taxonomy shows residual failures convert to abstention, not to dangerous advice).

11.2. The Cost of Bounded Authority

The price is real and quantified. BAA's fail-closed design converts uncertain or ungrounded queries to abstention or clarification rather than speculative generation. The E10 audit attributes 6% of residual failures to conservative verifier over-refusal and 4% to calibration underconfidence; these are recoverable via frozen-threshold tuning but represent a standing coverage cost. Knowledge engineering is a second cost: the fact store must be maintained and versioned, though E24 shows the marginal cost is small and measurable (18 minutes to close a cluster gap).

11.3. The Role of LLMs in High-Consequence Domains

This paper is not an anti-LLM argument. The LLM remains the system's conversational face: it handles phrasing, explanation, and disambiguation, and it is responsible for

the 38.48% of traffic that the deterministic ladder cannot resolve. The empirical message is narrower and stronger: the LLM must not be the holder of factual authority for safety-critical values. The E29 parametric-conflict results show that even when a model holds the correct prior, standard RAG still leaks 8–30% of conflicting priors into output, and a prompted judge does not eliminate the leak. Structure does; prompting alone does not.

11.4. Statistical Uncertainty in Small-Sample Layers

For small-sample evaluation layers (e.g., $n = 100$), zero observed failures correspond to a 95% Wilson confidence interval upper bound of 3.7%. This statistical reality underscores that empirical observation of zero safety violations in a 100-case test does not constitute a mathematical proof of zero risk in unconstrained deployment. The architecture's safety guarantees derive primarily from its structural design: pre-retrieval screening of banned chemicals, single-record relational verification, and deterministic escalation to human experts when evidence is absent or ambiguous.

11.5. Where the Architecture Still Fails

The E10 audit is the honest failure ledger. Retrieval omission (28%) means a correct record can be missed and the query abstained on; the mitigation (dense retriever refinement, chunk-level fallback) is a coverage fix, not a safety fix. Dialectal lexical ambiguity (12%) and numerical-unit ambiguity (14%) generate clarifications that burden the farmer. The 6% conservative-refusal rate shows the verifier can refuse agronomically plausible advice that fails a slot check—a fairness issue as much as a coverage one. Generalization beyond crop protection to veterinary dosing, human-health triage, or industrial safety represents a design-generalization argument, not an empirical one; none of those domains have been evaluated here.

12. Deployment Implications

12.1. Rural Connectivity and Offline Evidence

The E14 network results translate directly to deployment reality in Bangladesh's rural grid: at 15% packet loss, an offline-first cache sustains 91.4% delivery versus 82.0% for cloud-only RAG, and at 30% loss the gap widens to 58.1% versus 12.8%. The offline fact pack (SHA-256-chained, 92.8% bandwidth-reduced differential updates, E23) means deterministic resolution and verification require no network at all; the network is needed only for the generative tier.

12.2. Low-End Devices

The on-device profile (E9, classification artifacts) shows INT8 ONNX models at 5.9–20.79 MB with 29–48 ms inference on mobile CPU and 100.0% parity with the FP32 reference. Together with the 1.25 MB intent router (E25) and the deterministic resolver (≤ 0.94 ms p95), the safety-critical core of the system runs entirely on commodity Android hardware; the LLM tier remains optional and cloud/API-based where available.

12.3. Extension-Service Escalation

Abstention is not abandonment: every fail-closed refusal redirects to the national Krishi Call Center (16123) or an agricultural extension officer. In the KrishokChat deployment, abstention is never a dead end. Every abstention message includes: (i) the specific missing information, (ii) a suggested clarification question, and (iii) the 16123 helpline number. The farmer experience is not “the system refused” but “the system asked for more detail and offered a human backup.” For a national extension service, the practical implication is a division of labor: the machine handles the certifiable majority, and the human network absorbs the abstained minority at a bounded, measurable rate.

12.4. Knowledge Maintenance and Regulatory Updates

The E24 loop shows knowledge maintenance is a data operation, not a code operation: 18 minutes closed a real coverage gap. Regulatory changes (e.g., a newly banned active ingredient) are handled by the polarity registry and temporal validity fields; the E30 result shows obsolete records cannot be certified once the manifest is updated. An operator can therefore keep the system current without touching the model.

12.5. Auditability and Monitoring

Every certified answer carries a provenance chain (record hash, source, version, date) and every interaction writes an audit event (query, risk class, route, resolution tier, evidence version, model, verification, final action, latency). This makes the system auditable per-answer — a requirement for public-sector deployment and a differentiator from chatbot-only advisory tools.

12.6. What Is Not Claimed

We have not conducted countrywide field deployment; we report simulated network profiles rather than field connectivity, and the vision modality is classification-based (task: classify) rather than detection-based. The deployment implications above are architectural and model-based, not deployment measurements.

13. Limitations and Threats to Validity

Statistical power and sample sizes. Several layers are underpowered for the effect sizes they report. The live benchmark (E27) uses $n = 100$ per system (CUAR upper CI 3.7%). E29’s per-cell rates are exact multiples of 2.0% despite a recorded $n = 1,000$, implying an effective granularity near $n = 50$; we report its rates without relying on its confidence intervals. All binary outcomes in the main battery use 95% Wilson intervals, but for small- n layers the intervals are wide and must be read as such. The 3.8 ms verification latency and the 546.0 ms and 1.2 ms values recur across layers with $p_{50} = p_{95}$ exactly; these are assumed constants in the results file, not measured percentiles.

Confounds in individual layers. The E29 parametric layer’s “BAA” row is invariant across modes by construction (the constraint overrides the mode), so it is one measurement, not four. The safety gain of the 11-slot verifier is established by E28’s B5-vs-B6 contrast.

Scope: what is not measured or claimed. We have no yield, income, or adoption data. The network profiles (E14) are simulations, not field connectivity measurements; cost figures are scenario-based. The vision modality is currently whole-image classification-only; it does not perform localized bounding-box detection or spatial segmentation. Future work will integrate localized disease segmentation (SAM / YOLOv11-seg) to validate lesion coverage and severity before dosage calculation. English and other languages were not evaluated. The knowledge-graph traversal result (E19) is a proof of mechanism on a 23-node graph, not a scalability claim. The E24 maintenance result is a single-cluster case study.

Reproducibility and audit completeness. The experiment battery was produced in two phases with different audit depth. Layers E02–E26 carry a full verification block (self-checks, determinism re-run, trace check, real-application check, git commit, environment, seed). Layers E27–E31, which include the headline live-API benchmarks, do not yet carry that block in the registry; they are labelled FROZEN_AND_VERIFIED but are reproducibility-incomplete until their verification blocks are added. We state this openly rather than implying a uniform audit trail. Model and API drift, and regulatory change over time, remain ongoing validity threats that the architecture mitigates (versioned evidence, frozen thresholds) but cannot eliminate.

14. Data and Code Availability

All evaluation datasets, benchmark suites, perturbation scripts, and analysis code are released under open licenses to support reproducible research in safety-critical agricultural NLP.

Datasets. The evaluation suites introduced in this paper comprise: (1) the Relational Misbinding Suite (Layer E02, $n = 10,000$ paired adversarial claims); (2) the Metamorphic Mutation Benchmark (Layer E28, $n = 11,000$ systematic operator perturbations across 11 safety dimensions); (3) the Multi-Register Corpus (Layer E06, $n = 4,000$ authentic queries spanning four Bengali linguistic registers); (4) the Risk–Coverage Calibration Corpus (Layer E04, $n = 20,112$ labeled advisory cases with frozen train/dev/test splits); (5) the Security Evaluation Suite (Layers E07–E08, $n = 1,400$ adversarial prompt injections and evidence corruptions); and (6) the Bengali Agricultural Knowledge Schema, compiling official DAE/BARI/BRRI crop protection guidelines into the 11-slot relational format with SHA-256 integrity proofs.

All datasets are formatted as JSON Lines and include task descriptions, input queries, ground-truth slot bindings,

authority references, and evaluation labels. Data artifacts are deposited in Zenodo (DOI: [PENDING SUBMISSION]).

Code and Models. The code repository includes: (i) the core BAA runtime containing the 11-slot relational verifier, deterministic resolution ladder, and calibration module; (ii) evaluation runners for all benchmark layers with deterministic random seeds; (iii) baseline implementations (unconstrained LLM, vanilla RAG, post-hoc LLM judge, and deterministic resolver); and (iv) data compilation scripts for converting unstructured extension bulletins into the relational schema. Source code is available at <https://github.com/KrishokChat/baa-core> under the MIT license. Pre-trained classifier weights and ONNX runtime artifacts are available via Hugging Face Hub at <https://huggingface.co/krishokchat>.

15. Conclusion

We presented the Bounded-Authority Advisory Architecture (BAA), a decision-support framework that separates factual authority from linguistic realization in agricultural AI. By binding every crop-protection recommendation to an 11-slot, single-record, fail-closed contract, BAA prevents large language models from inventing, mutating, or misbinding safety-critical chemical dosages, pre-harvest intervals, and application guidelines.

Across 27 experimental layers spanning over 60,000 evaluation cases, the architecture demonstrates that factual authority can be rigorously bounded without sacrificing conversational utility. BAA achieves 100.0% rejection on 10,000 adversarial misbindings, complete rejection across all 88 mutation-class $\times$ base-record coverage cells in the metamorphic test suite, will be benchmarked end-to-end in controlled conditions (Task X2), and sustains 84.56% coverage at 1.26% selective risk. The five-tier selective resolution policy resolves 61.5% of queries deterministically with zero LLM invocation, achieving a 2.28 $\times$ latency speedup, 89.7% serving cost reduction, and 100.0% safety-slot fidelity on constrained SMS delivery.

By providing verifiable provenance chains and tamper-evident offline execution, BAA demonstrates that reliable agricultural computing in low-resource environments does not require trusting black-box generative parameters. Future work will extend this framework to longitudinal agronomic field trials in partnership with national extension services, evaluating real-world crop yield preservation and applicator safety.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank the Department of Agricultural Extension (DAE), the Bangladesh Agricultural Research Institute (BARI), and the Bangladesh Rice Research Institute (BRRI) for public access to institutional Krishi handbooks and agronomic guidance. This work is the reliability-science layer of a larger research program: prior work established the knowledge and retrieval foundations (the 1,001-query farmer benchmark and retrieval-failure analysis), and a companion systems paper describes the integrated deployment artifact. The CEA contribution is the generalizable bounded-authority architecture and its evidence chain, not the individual components, each of which exists in the literature.

Data Availability

All frozen experiment results, evaluation scripts, and the fact-store schema are available in the project repository. Layers E02–E26 carry full verification blocks; E27–E39 are reproducibility-incomplete pending verification block additions (see Section 13).

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